



Lab 12

MOSFET Current Source and Current Mirrors

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12.1 Objective

To design MOSFET current

source and current mirror circuits and utilize current-mirror circuit for biasing of MOSFET CS amplifier.

Task 1: Pre-Lab

Q1. Why do we use MOSFETs as a Current Source, Explain?

A MOSFET operating in the saturation region (also called active region) exhibits a unique characteristic: the drain current (I_D) becomes relatively independent of the drain-source voltage (V_{DS}) and depends primarily on the gate-source voltage (V_{GS}). This property makes MOSFETs ideal for current source applications. They provide a wide voltage compliance range over which the constant current is maintained, allowing flexible circuit design.

Advantages of MOSFET Current Sources

Voltage Compliance: Wide range of output voltage over which constant current is maintained

Tunability: Current magnitude easily adjusted via W/L ratio or V_{GS}

Integration: Compatible with standard IC fabrication processes

Low Power: Negligible gate current in steady-state

Scalability: Multiple current sources easily implemented on same chip

Q2. What is MOSFET Current Mirror? What is the significance of using Current Mirrors?

A current mirror is a circuit designed to replicate a reference current (I_{ref}) into one or more output branches with high accuracy. The basic two-transistor MOSFET current mirror consists of:

Input transistor (M1): Diode-connected, establishes reference current. Output transistor (M2): Mirror transistor that replicates the current

Fundamental Significance

Current Replication: Provides multiple identical or scaled current sources from single reference

Essential for biasing multiple stages in analog circuits

High Output Impedance: Acts as near-ideal current source. Minimizes loading effects on connected circuits

Temperature and Supply Tracking: All mirrored currents track variations together. Provides inherent compensation in differential circuits